Exp. No: 3

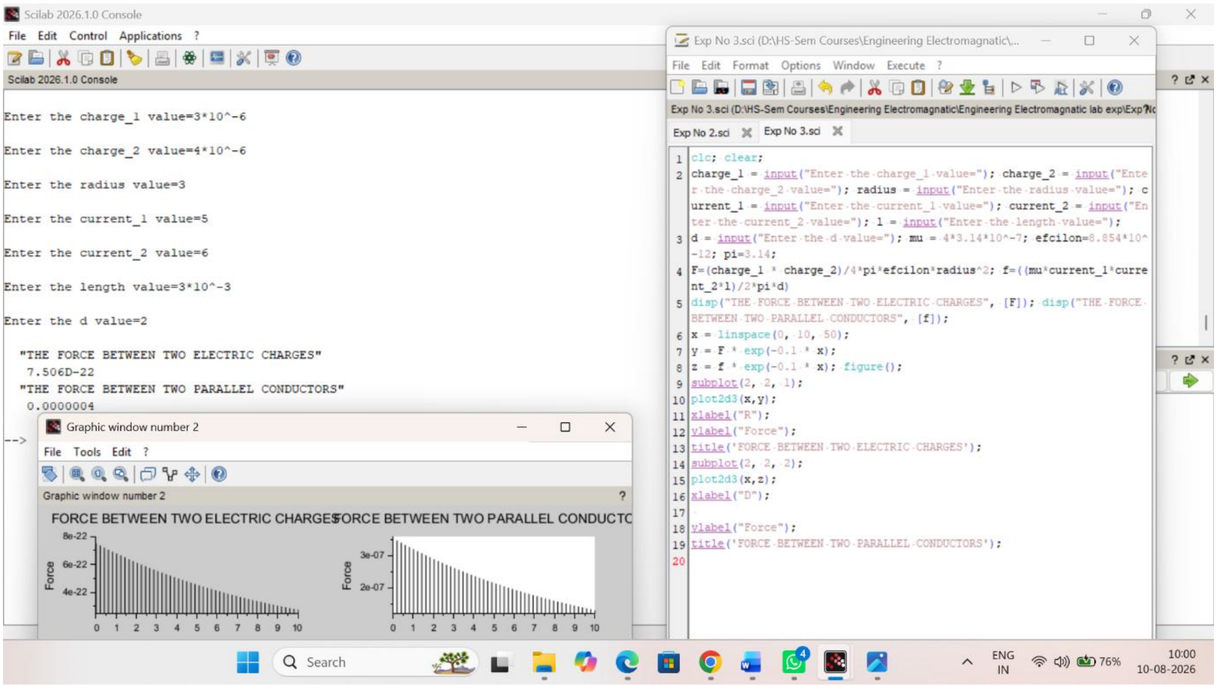
Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required:

• SCILAB version 6.0.1



EXP NO :- 03

$q_1 = 3 \times 10^{-6}$

$q_2 = 4 \times 10^{-6}$

$r = 3cm$

$I_1 = 5$

$I_2 = 6$

$l = 3mm$

Force B/w two electric Charg

$$F = \frac{K q_1 q_2}{r^2} = \frac{8.99 \times 10^9 \times (3 \times 10^{-6})(4 \times 10^{-6})}{3^2}$$

$$= 8.99 \times 10^9 \times (1.333) \times 10^{-12}$$

$$\boxed{F = 0.01199 N}$$

Force B/w two/parallel Conductor

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$= \frac{(4\pi \times 10^{-7})(5)(6)(3 \times 10^{-3})}{2\pi(8)}$$

$$\frac{(2 \times 10^{-7}) \times 30 \times 3 \times 10^{-3}}{4\pi}$$

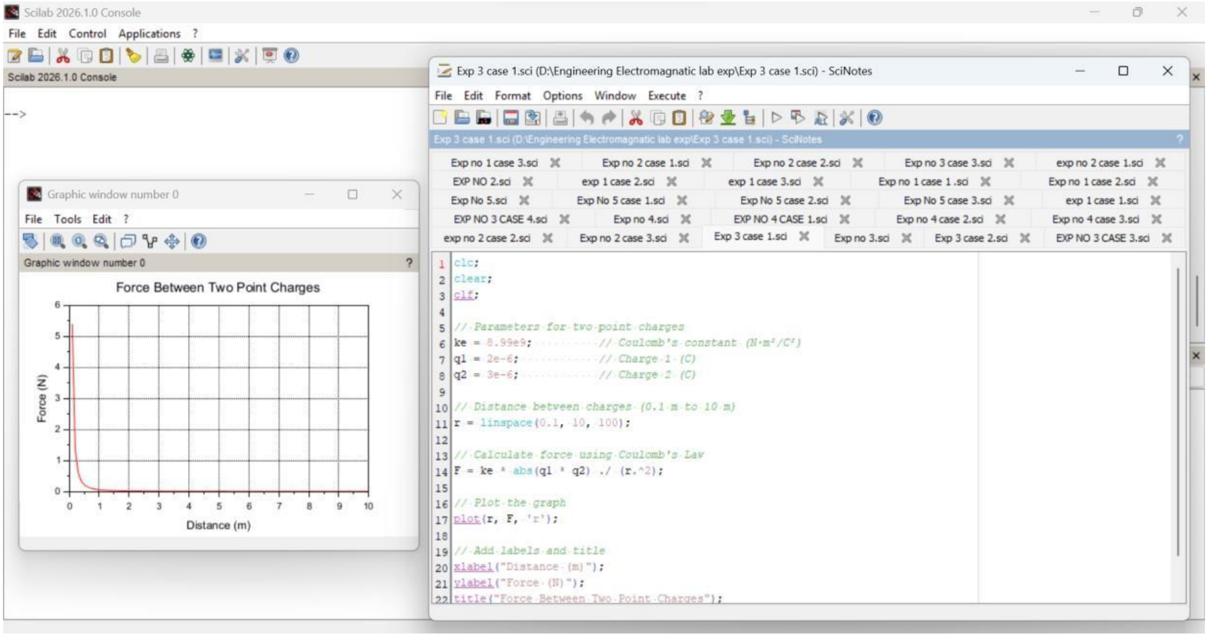
$$\frac{1.8 \times 10^{-8}}{4}$$

$$\boxed{F = 4.5 \times 10^{-5} N}$$

O/P ✓

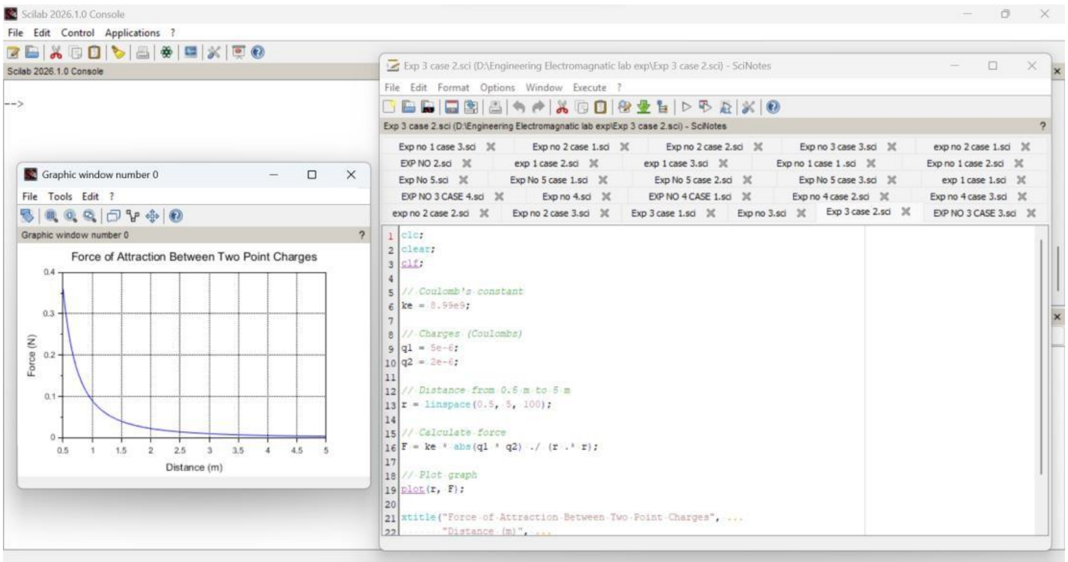
Case 1:

Given two-point charges q1=2×10^−6 and q2=3×10^−6 C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.



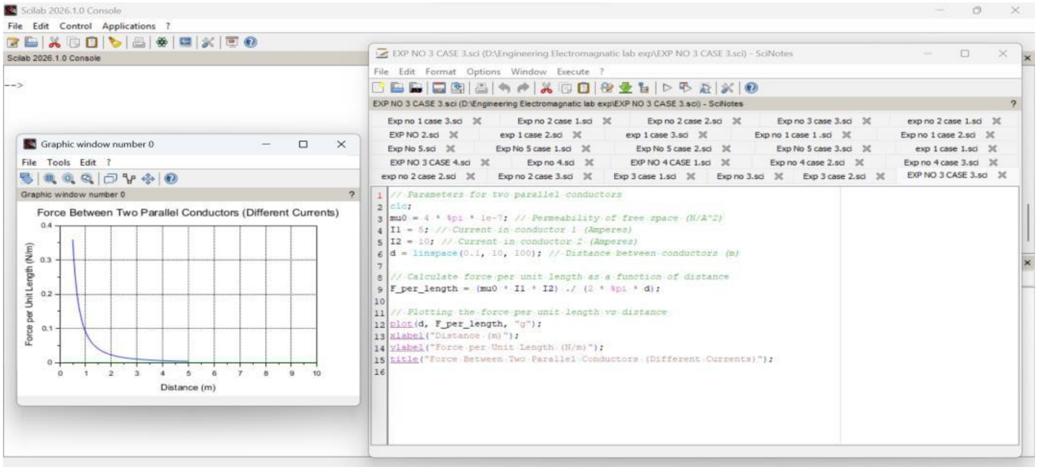
Case 2:

Given two-point charges q1=5×10^−6 C and q2=2×10^−6, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.



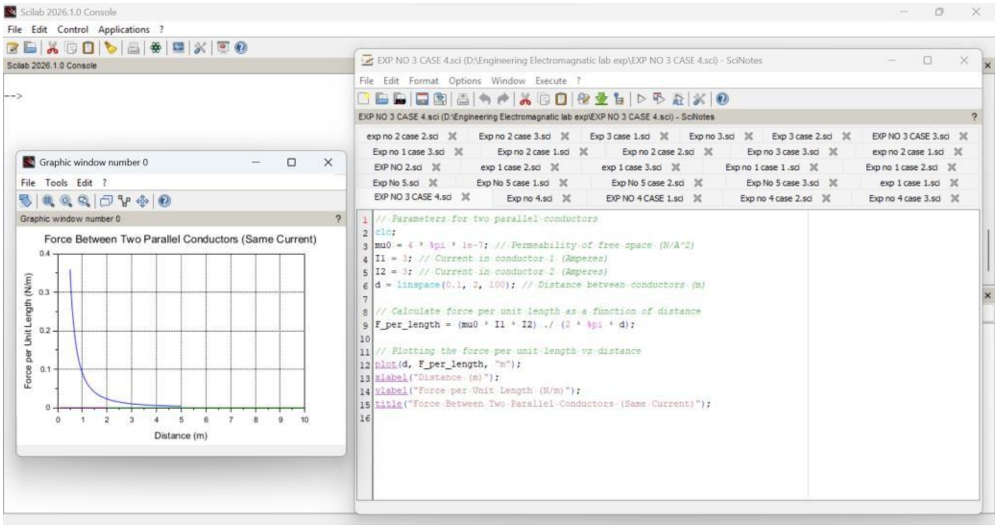
Case 3:

Two parallel conductors are separated by a distance d. The first conductor carries a current of I1=5 A, and the second conductor carries a current of I2=10 A. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors



Case 4:

Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.



Result : Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.